

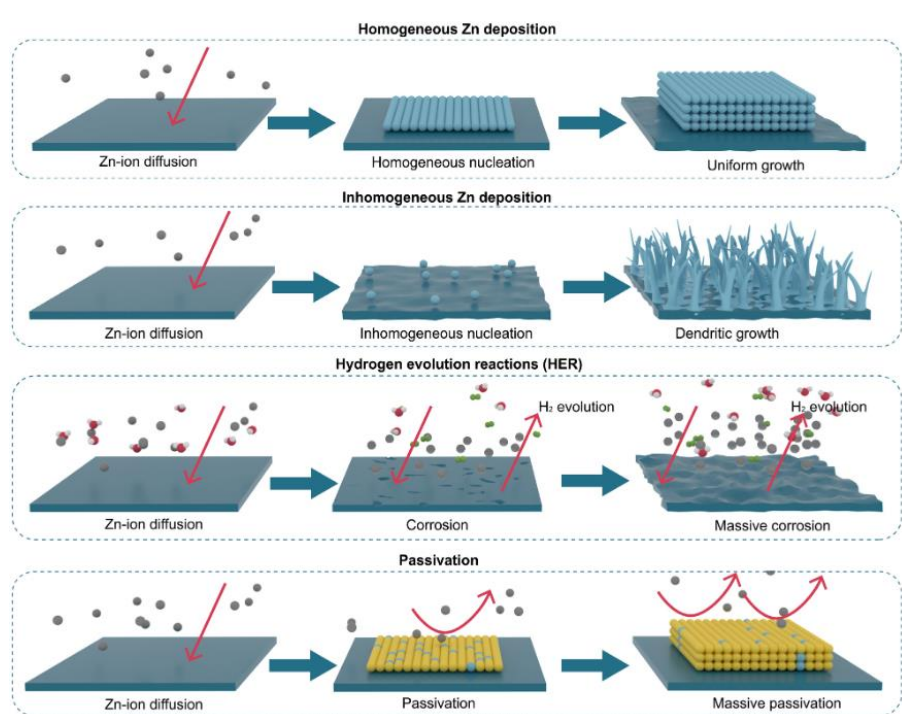
HIGHLY REVERSIBLE ZN ANODE WITH A PRACTICAL AREAL CAPACITY ENABLED BY A SUSTAINABLE ELECTROLYTE AND SUPERACID INTERFACIAL CHEMISTRY



Chang Li, Abhinandan Shyamsunder, Alexis Grace Hoane, Daniel M. Long, Chun Yuen Kwok, Paul G. Kotula, Kevin R. Zavadil, Andrew A. Gewirth, and Linda F. Nazar

Motivation

- Aqueous zinc-metal batteries (AZMBs) are plagued by poor Zn reversibility owing to zinc dendrite and side reactions.
- In this work, we developed a low-cost electrolyte with a specialized additive to achieve highly reversible Zn anode



Zn anode performance

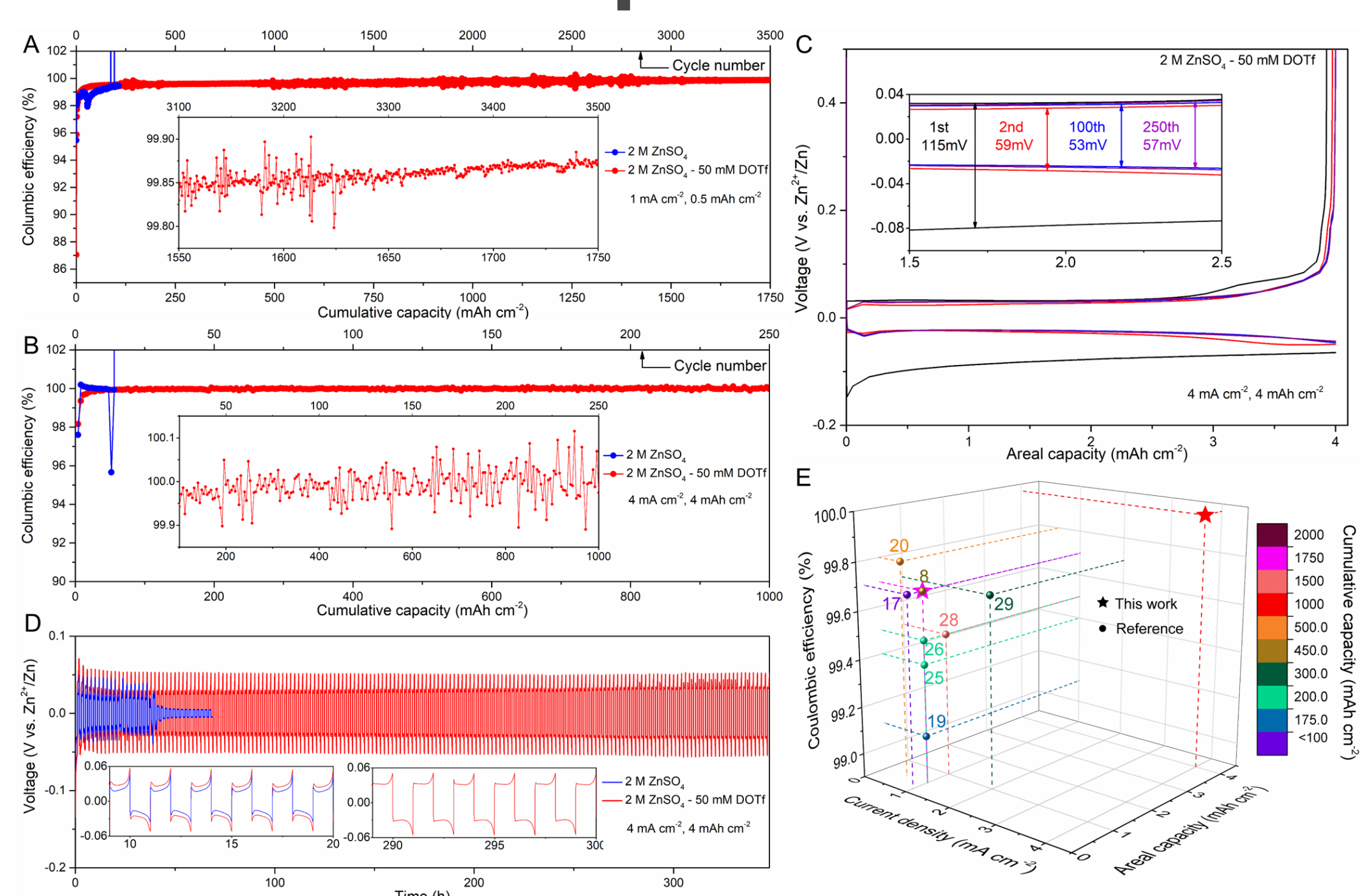


Figure 1. Highly Reversible Zn Plating/Stripping

- 50 mM N,N-dimethylformamidium trifluoromethanesulfonate (denoted as DOTf) was used as the additive for aqueous 2 M ZnSO_4
- Highly stable and reversible Zn plating/stripping at both a lab-level capacity of 0.5 mAh cm^{-2} and practical-level capacity of 4 mAh cm^{-2} is achieved with DOTf additive. (Figure 1A-D)
- The modified electrolyte demonstrate excellent performance compared to other reported Zn electrolytes, considering four important parameters: Zn plating/stripping coulombic efficiency (CE), current density, areal capacity of each plating and stripping, as well as the total cumulative capacity. (Figure 1E)

Zn deposition morphologies

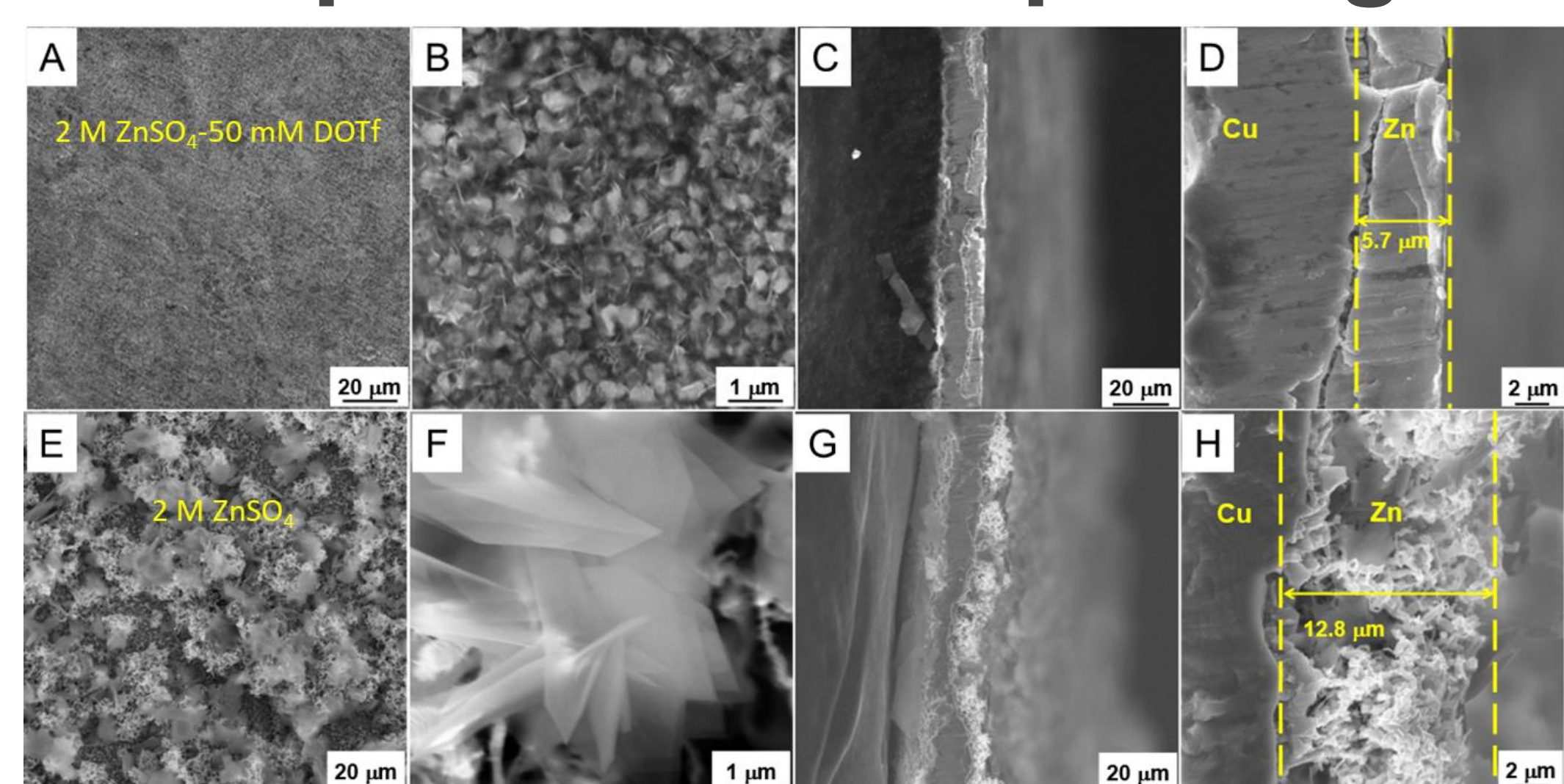


Figure 2. Morphologies of plated Zn on Cu

- Plated Zn on Cu electrodes after the 10th plating at 4 mA cm^{-2} and 4 mAh cm^{-2}
- DOTf enables homogeneous, smooth and compact Zn deposition layer (Figure 2A-D).
- Porous and dendritic Zn was observed in baseline electrolyte (Figure 2E-H).

Solid-electrolyte-interface

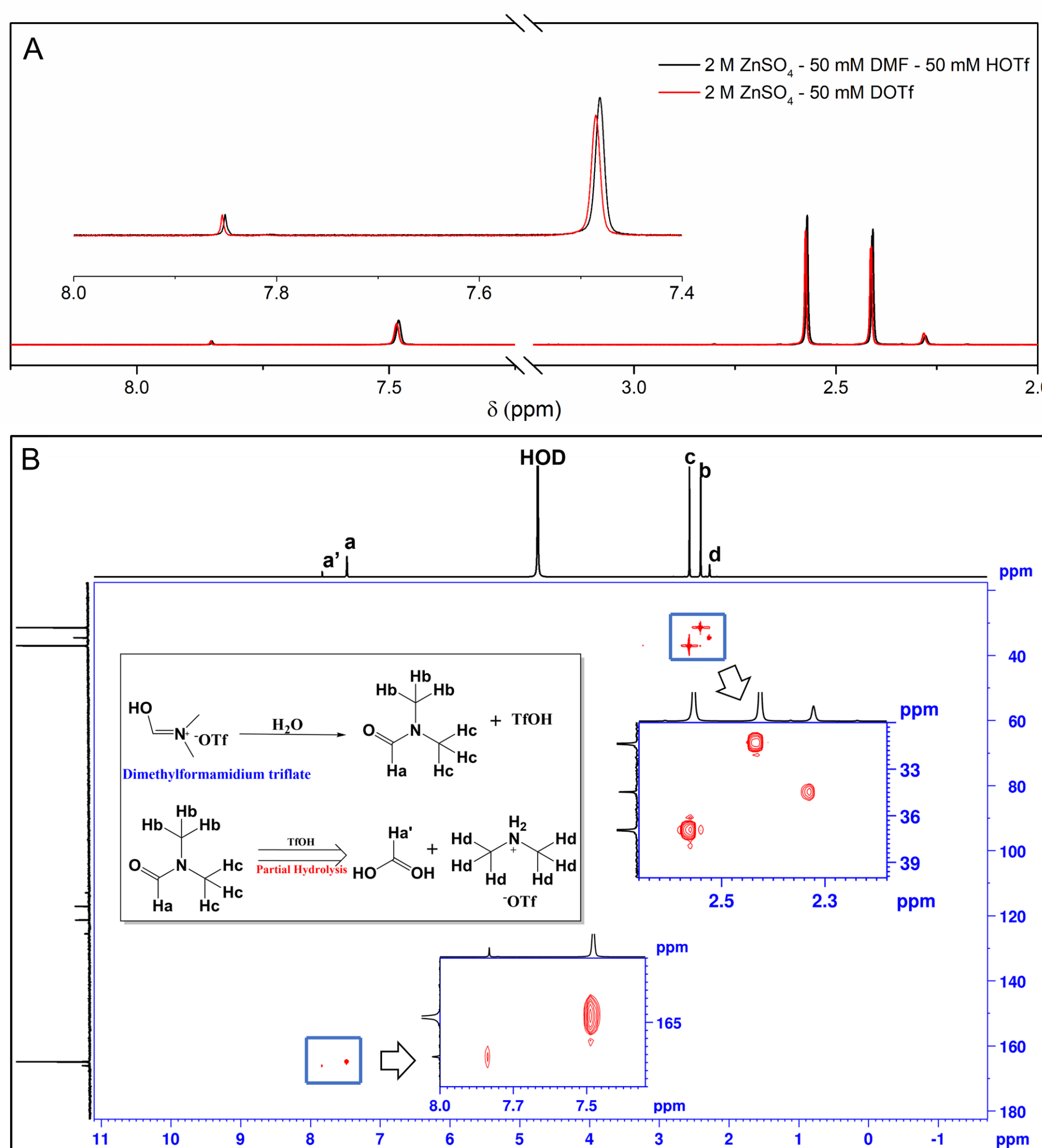


Figure 3. NMR studies for the 2 M ZnSO_4 - 50 mM DOTf Electrolyte

- DOTf exists as its "neutral ion pair" in aqueous electrolytes: $[\text{DMF}]^+[\text{OTf}]^- \rightarrow \text{DMF} + \text{HOTf}$

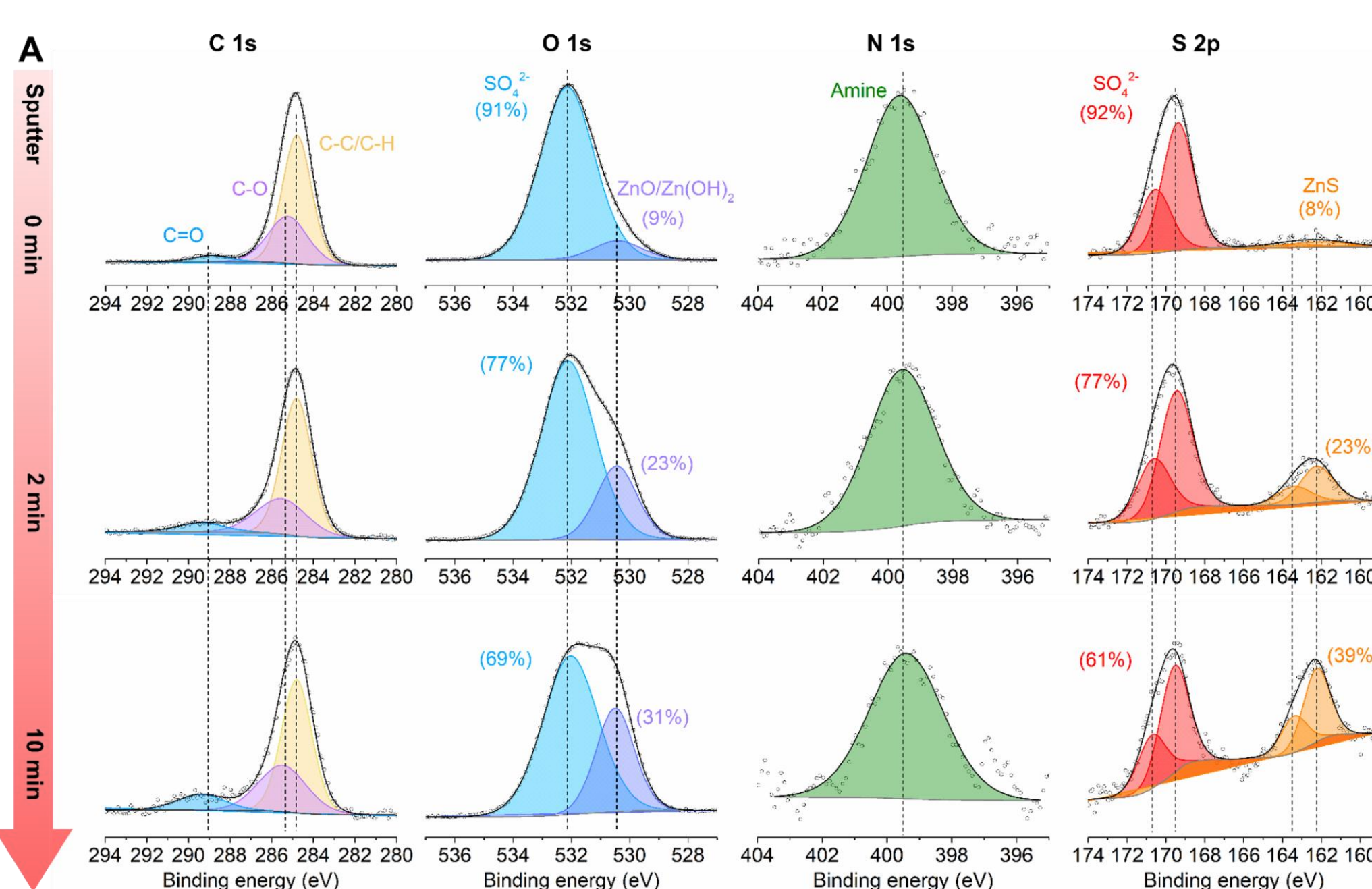


Figure 4. Interfacial studies of the Zn/electrolyte interface

- A complex solid electrolyte interface SEI (including oxo, amine, sulfide) on surface of deposited Zn

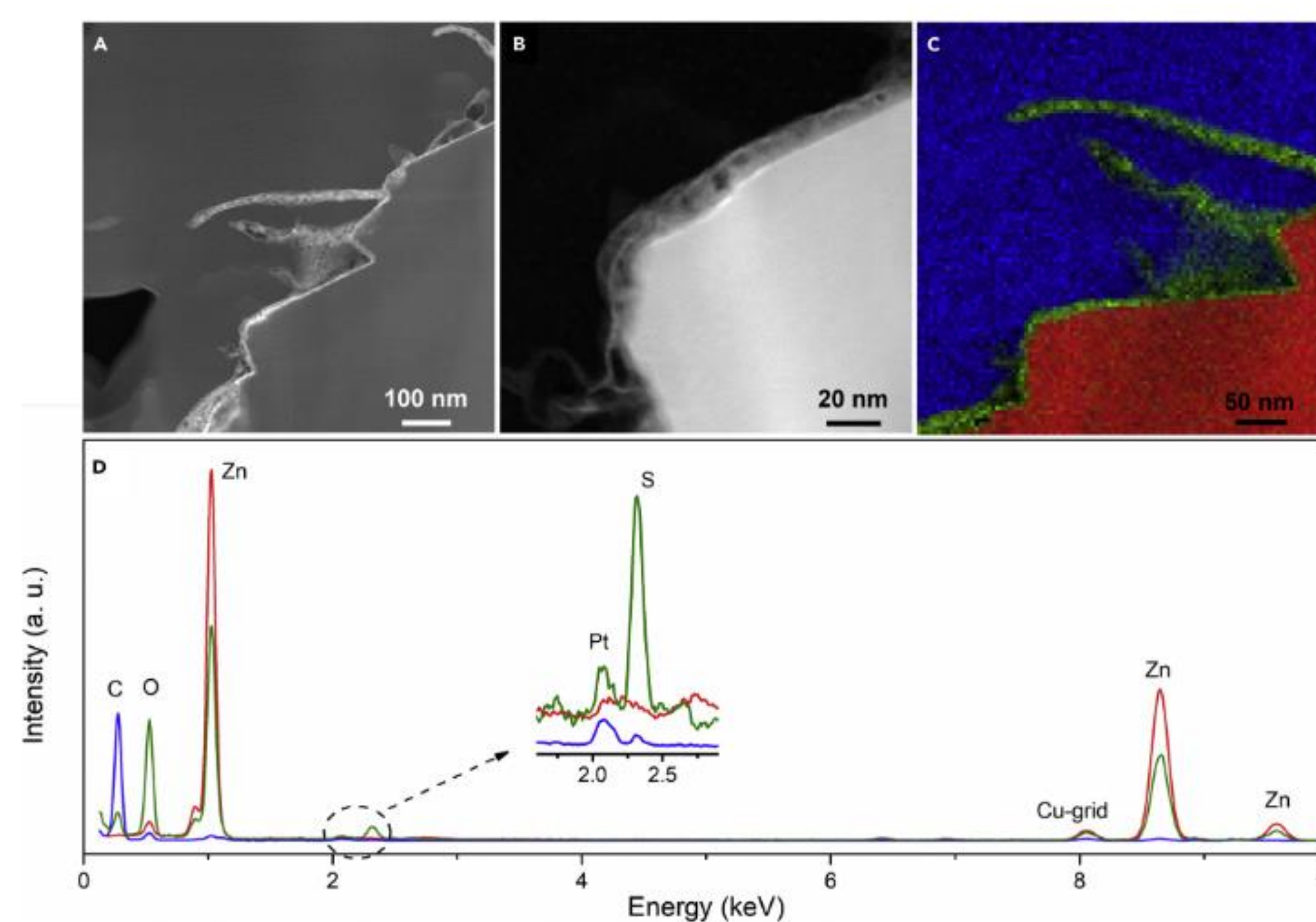


Figure 5. Interfacial studies of the Zn/electrolyte interface

- A nanometric, porous SEI is responsible for facilitating Zn cation transport during plating/stripping

Nanoscale Zn nucleation

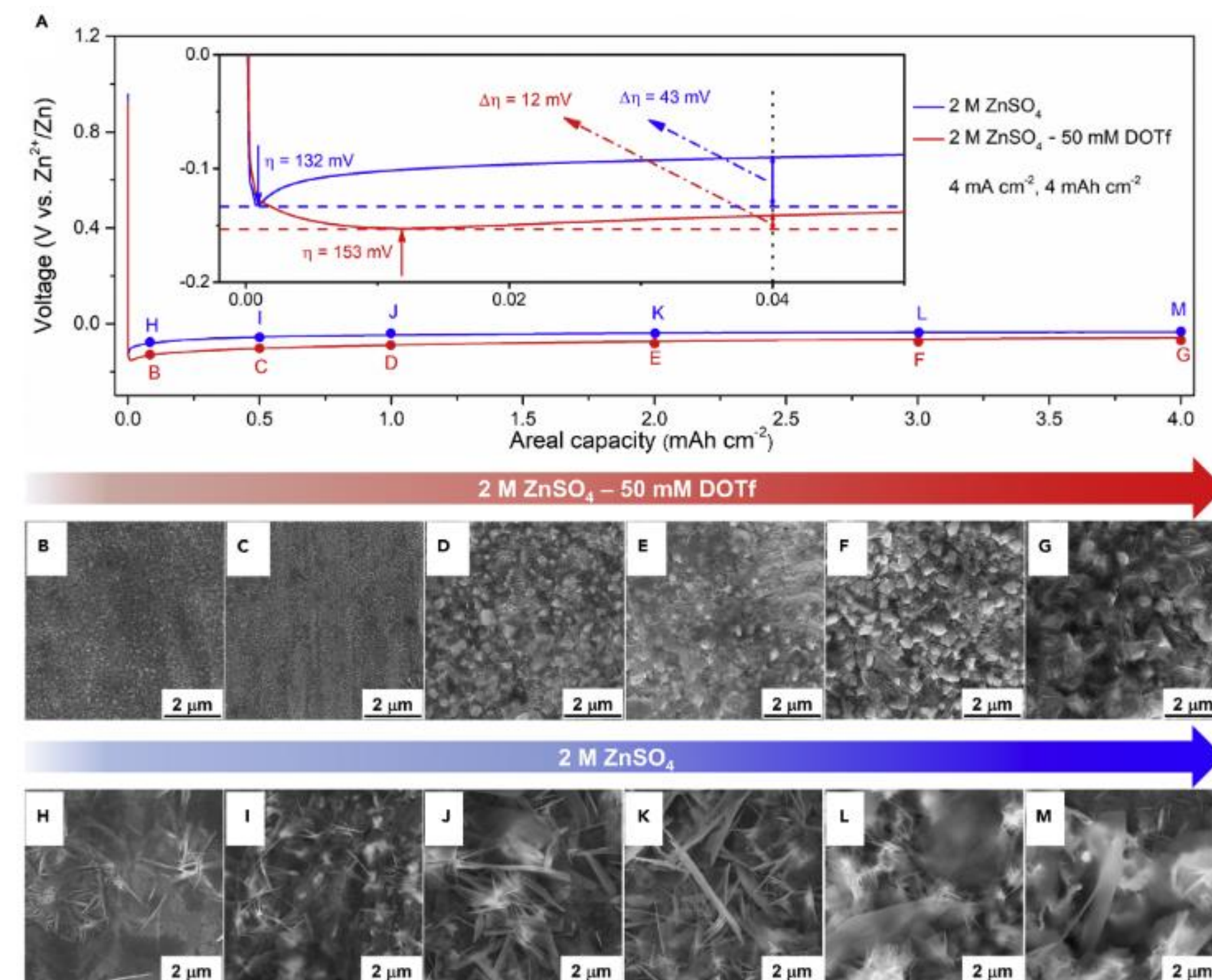


Figure 6. Nanoscale nucleation and growth of plated Zn

- The SEI manipulates the initial nanoscale Zn nucleation and growth behavior

Full cell performance

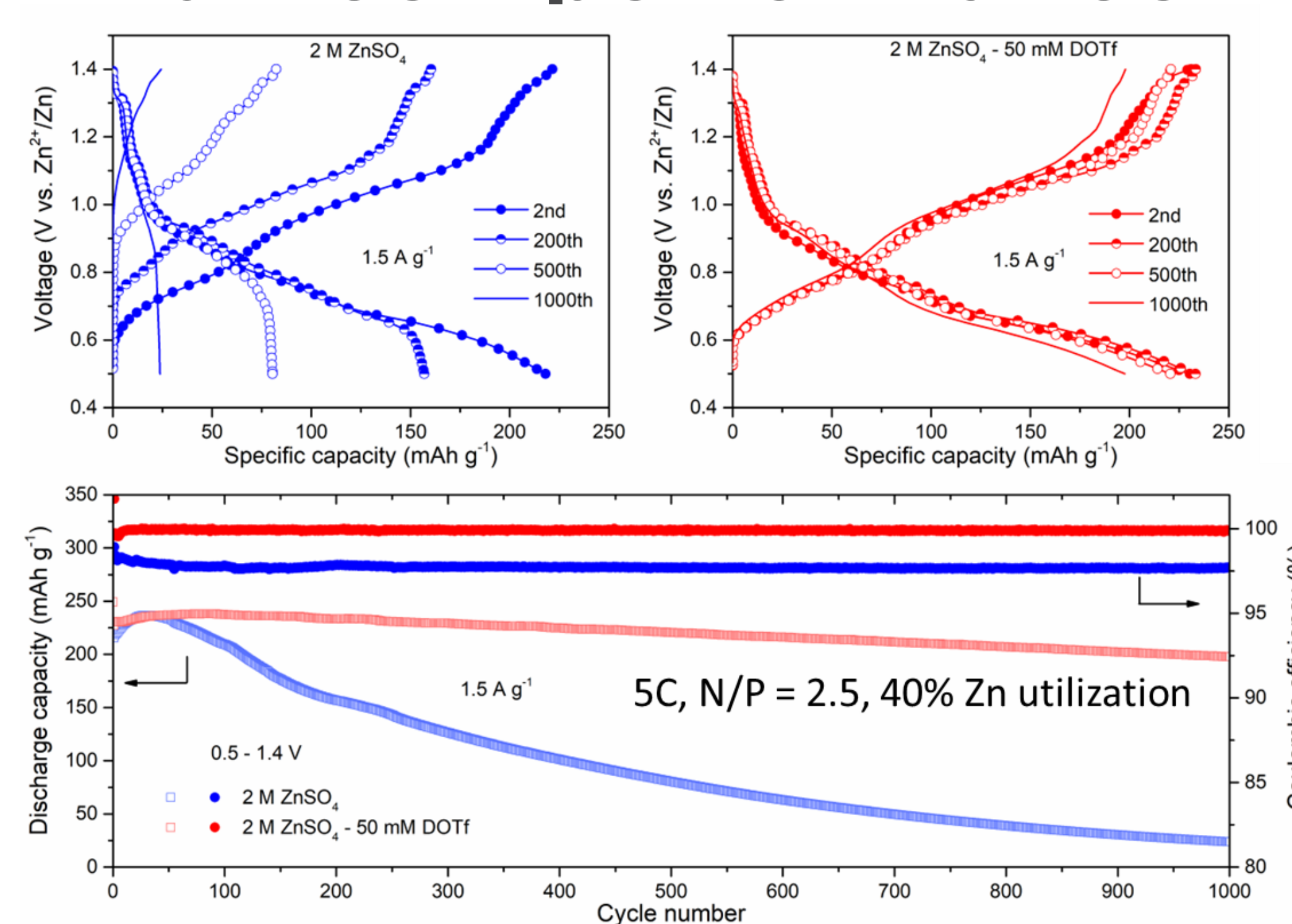


Figure 7. Electrochemistry of $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ full cells

- DOTf-modified electrolyte enables excellent performance of $\text{Zn}||\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ cells under high Zn utilization conditions

Summary

- In this work, we introduce DOTf as a novel electrolyte additive for AZMBs for dendrite-free and highly reversible Zn plating/stripping.
- The origin of this outstanding performance is a result of the water assisted dissociation of DOTf into neutral ion pairs and resulted SEI layer on Zn.

References

- [1] Li, C. et al. Highly Reversible Zn Anode with A Practical Areal Capacity Enabled by A Sustainable Electrolyte and Superacid Interfacial Chemistry. *Joule* **6**, 1103-1120 (2022).
- [2] Liu, Y. et al. Rechargeable aqueous Zn-based energy storage devices. *Joule* **5**, 2845-2903 (2021).

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